As the URLs are SSL enabled, allow the browser to process further for the first time. Enter 'root' and its password to log in. On logging in, a warning message for the subscription is displayed. Click on *OK*, since there is no valid support subscription from Proxmox yet, as it is just being set up.

Creating a cluster and a joining cluster

After logging in to both servers, first create a cluster on the pve01 server and join the pve02 server to it. This allows the management of both servers from any of the server panels and helps in building replication.

- Now on the first server, click on *Datacenter* in the navigation menu on the left.
- The right-side displays a list sub-menu, in which you can click on *Cluster*.
- Click on the *Create Cluster* button, enter the cluster name as *pvecluster* and click on *Create*.
- After the cluster is created, copy the joining encrypt key by clicking on *Join Information*. A popup window opens with the encrypt key. Click the *Copy information* button —this copies the code to the clipboard for pasting into the second server.
- Now on the second server, click on *Datacenter,* which is on the left side menu and go to a similar cluster section. Click on the *Join Cluster* button and paste the join information from the clipboard. The peer address and fingerprint section is auto-filled. Give the root password of the first server, and this then joins the second server to the first server.
- *pve01* and *pve02* will be seen on both servers' Web-admin panel. You can close the admin panel of the second server pve02, as all activities will be carried out on the pve01 admin panel.

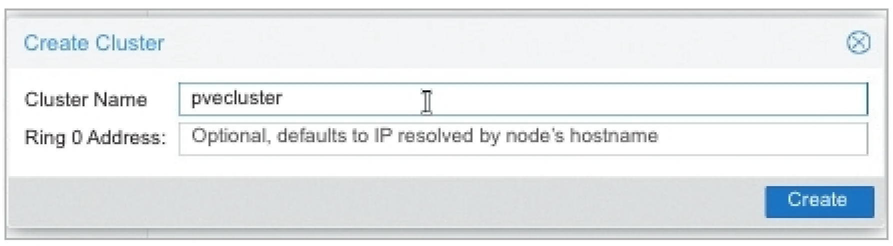

Figure 1: Creating a cluster

Create storage with the ZFS file system for replication between the servers

Let's create the ZFS file system on both servers for a 600GB HDD, as this will be used for storing VM images.

- Click on *pve01* from the left side navigation menu; a sub-menu on the right-side panel for the given server will show up.
- Clicking on *Disk* opens a sub-level menu. Scroll down and click on *ZFS*.
- Now, from the top menu, click on the *Create ZFS* button, and a pop-up window will ask for details.
- Enter the name as *zfsdata,* then select *Single Disk for RAID,* and leave the rest of the settings as they are. Select the 600GB HDD shown in the *Device* section, which is usually called */dev/sdb*.
- After selection, click on the *Create* button. This creates storage on pve01. Follow the same process on the pve02 server to add ZFS with the same name, *zfsdata*.
- As ZFS storage is created with the same name *zfsdata*, it is now available for replication between the servers.

Uploading the ISO for the installation of VMs

As we are going to create various VMs, we need to first download the ISO and upload on pve01 local storage for use. We will create two VMs: one based on Windows 10 and another on the Ubuntu 18.04 Linux desktop. For download, refer to the links given below (ensure that you have an official Windows 10 licence-key ready):

- *https://www.microsoft.com/en-in/software-download/windows10ISO*
- *http://releases.ubuntu.com/18.04.1/ubuntu-18.04.1-desktop-amd64.iso*

Now we are going to upload this into a Proxmox cluster:

- Click on the down arrow before *pve01* in the left-side navigation menu; this expands *pve01* further and *local(pve01)* will be visible.
- Click on this storage local on the right-side and an option called *Content* will open up.
- Click on *Content*, and then on the *Upload* button, which will open a dialogue box. Select *ISO Image* and then choose *Download ISO files* one by one, and upload them here. Both Windows and Ubuntu ISOs are now shown in the *Contents* section.

Creating Windows and Linux VMs

Creating a VM is quite simple and straightforward.

- Click on the *Create VM* option given on the top-rightmost corner. It is highlighted in bright blue.
- Its first tab is general, asking you to give a name for the VM. For MS Windows, let the name be Windows10VM and when installing Ubuntu Linux, name it Ubuntu18VM. The remaining settings are auto generated, so let them be as they are.
- Next go to the *OS* tab and select the ISO image based on the operating system installation; in the *Guest OS* section, select *Type as Microsoft Windows* with 10/2016 in the *Version* drop-down menu. In Ubuntu Linux, we will select *Type as Linux* and choose 4.x/3.6/2.6 kernel

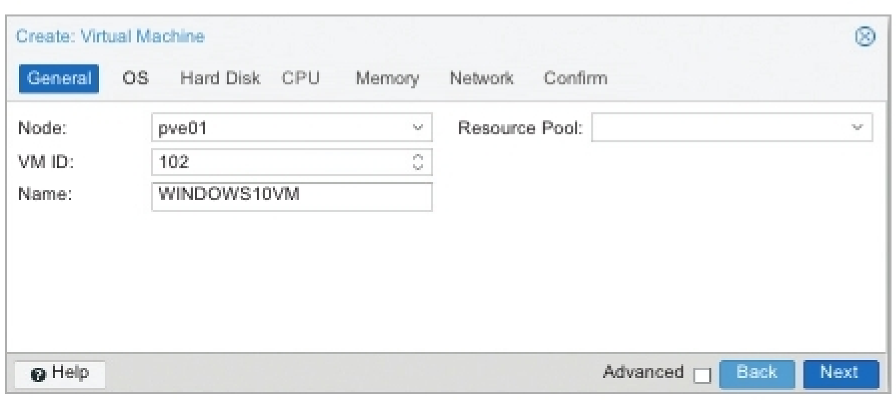

Figure 2: Creating VM